Nanocellular Foams of PS/PMMA Polymer Blends

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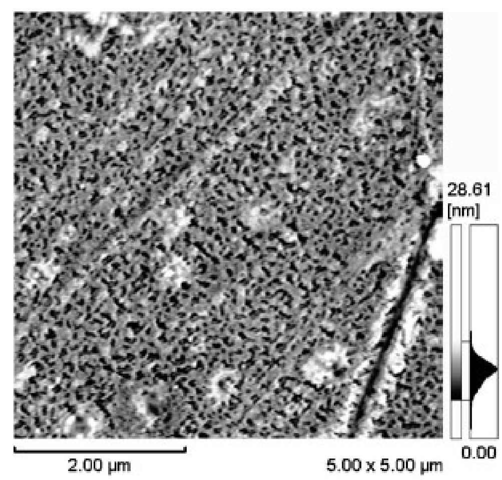
A nanocellular PS/PMMA polymer blend foam was prepared, where bubble nucleation was localized in the PMMA domains. The blend, which contains dispersed nanoscale PMMA islands, was prepared by polymerizing MMA monomers in a PS matrix to form highly dispersed PMMA domains in the PS matrix by diffusion mixing. The resulting blend was foamed with CO_2 at room temperature. A higher depressurization rate at lower foaming temperature made the bubble diameter smaller and the bubble density larger, and a higher PS composition in the blend resulted in a larger bubble density. A void with 40–50 nm in average diameter and a pore density of 8.5×10^{14} cm^{-3} was obtained as for the finest nanocellular foams.



Introduction

Many studies on creating high-performance polymeric foams have focused on increasing the number of cells and decreasing cell size. Microcellular foaming is one technique that has been studied for this purpose. Development of nanocellular foams now attracts great attention in research conducted on foam. Rapid pressure quenching of high-T_g polymers,[1] stress-induced foaming[2] and nanocomposites[3] methods could successfully prepare 1-micron or submicron size cells in polymer matrices. However, to the best of our knowledge, no foam possessing cells smaller than 100 nm has been reported.

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There were some studies on the preparation of nanocellular foams with self-assembled block copolymers. Recently, Yokoyama et al.,[4,5] Li et al.[6] and Taki et al.[7] reported the preparation of a nanocellular foamed thin film with 10-nm size cells using a diblock copolymer as a templating material and carbon dioxide as the foaming agent. The concept of their nanocellular foaming was to use the diblock copolymer possessing nanoscale-ordered spherical morphology as a template for bubble nucleation and growth. The block polymer of Yokoyama's group was a polystyrene-*block*-poly(perfluorooctylethyl methacrylate) (PS-*b*-PFMA) diblock copolymer. It creates a spherical morphology with PFMA as a spherical microdomain. Since PFMA contains fluorine, the PFMA domain is considered to be CO_2-philic, and the dissolved CO_2 is believed to localize in the spherical domains of PFMA. The difference in CO_2 concentrations between two domains, PS and PFMA, confines nucleation to the PFMA domain and suppresses

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bubble coalescence by the surrounding PS domain. In their method, cellular size was controlled by using the phase morphology of the block copolymer as a template. However, in their papers, nanocellular foaming was based only on the difference in the affinity of the polymers for the foaming agent (CO_2). That is, using the fact that CO_2 concentration was higher in PFMA than in PS, bubble nucleation in the polymer was confined in the CO_2-philic, PFMA, domain. However, an important factor for creating nanocellular structure, the viscoelasticity differences between the disperse domain and the matrix, was missing from their proposed mechanism. In order to make the principle of nanocellular foaming to be more general, the viscoelasticity differences between disperse and matrix polymers should be considered and intentionally designed to control bubble location. Furthermore, the morphology of block copolymer is very sensitive. It is easily affected by molecular weight distribution, surface energy between substrates and polymer, and the thickness of the film. Therefore, it would be very difficult to use the morphology of self-assembled block copolymers for creating the nanocellular foams as industrial products.

In this study, extending the principle of nanocellular foaming, i.e., localizing bubble nucleation in a confined domain, a nanocellular foam was prepared with a blend of polymers, polystyrene (PS) and poly(methyl methacrylate) (PMMA). The polymer blend was prepared by polymerizing methyl methacrylate (MMA) monomers in a PS matrix after dissolving the MMA monomer into the PS matrix. The polymerization in the polymer matrix provided highly dispersed PMMA domains. The polymer blend was then foamed by CO_2. The effects of blend ratio, foaming temperature and depressurization rate on bubble diameter as well as bubble density were investigated.

Experimental Part

PS ($\overline{M}_w = 230\,000$), was purchased from Aldrich Chemical Co. Ltd. It was provided in pellet form and used without further purification. MMA monomer, 98.0% in purity (Wako Pure Chemical, Japan), was purified by a rotary-evaporator and used as a solvent as well as a monomer. 2,2′-Azoisobutyronitrile (AIBN, 98%, Wako Pure Chemical, Japan) was used as an initiator of PMMA polymerization, and CO_2 (99.9%, Showa Tansan) was used as a physical blowing agent. In order to prepare nanoscale-dispersed PMMA domains, the method called polymerization in polymer was developed.

The procedure is illustrated in Figure 1. At first, the MMA in liquid form was mixed with 1 wt.-% AIBN. The PS pellets were then immersed into the MMA solution to let MMA monomers diffuse into the pellets. The mixing was conducted in an ample tube at a low temperature, 5 °C. After 24 h mixing time, a homogeneous PS/MMA solution was obtained and poured into a disk shape mold 1 mm in thick and 2.5 cm in diameter. The mold was heated up to 80 °C for 1 h to polymerize the MMA monomers in the PS matrix with AIBN. After polymerization, the molded samples were placed in a temperature-controlled incubator and kept at 90 °C for 24 h. Then, the samples were moved into a vacuum dryer and annealed at 170 °C in a vacuum for 24 h to completely polymerize the remaining MMA monomers in the PS matrix. The molecular weight of the PMMA/PS blend was measured by GPC (RID-10A/SPD-20A, Shimadzu, Japan). The degree of dispersion of the PMMA domain was checked by observing the morphology of samples by scanning electron microscopy (Tiny-SEM, Technex, Japan) after removing the PMMA domain with acetic acid.

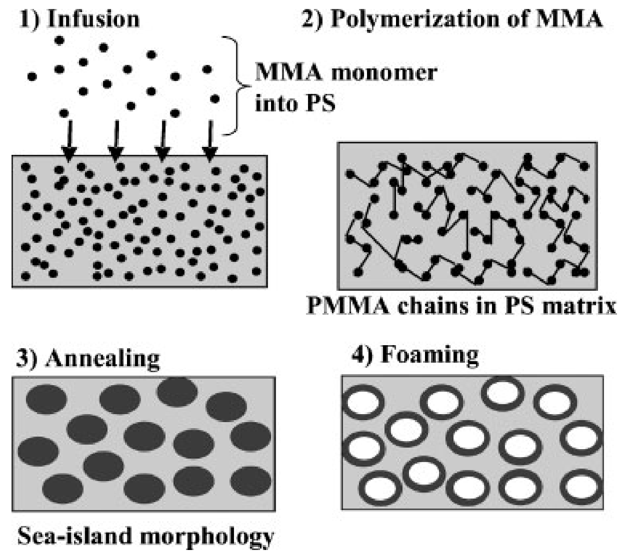

Figure 1. Preparation of nanoscale dispersed PMMA in PS matrix and idealistic nanocellular foaming.

The annealed sample was foamed by a pressure quench method with CO_2 after it cooled down to room temperature. To observe the effect of morphology as well as foaming condition on cell structure, the samples were prepared at three different PS and PMMA weight ratios, PS/PMMA = 3/1, 2/1, and 1/1 by varying the weight ratio of the PS pellet and MMA monomers through a mixing process. Furthermore, the foaming experiments were conducted at different foaming temperatures and depressurization rates.

The annealed sample was first put into the high pressure autoclave, heated up to 40 °C and pressurized by CO_2 up to 8.6 MPa. After keeping the sample under pressurized CO_2 for 2 h to dissolve CO_2 into the sample, the temperature was changed to a specified foaming temperature while keeping the CO_2 pressure at

Table 1. Batch Foaming condition.

Condition	Temperature	Depressurization rate
	°C	$MPa \cdot min^{-1}$
A	40	0.1
B	20	0.1
C	20	0.5
D	40	0.5

8.6 MPa. When the temperature had reached the given foaming temperature, the pressure was reduced to 0.1 MPa, atmospheric pressure, at the given depressurization rate by releasing the CO_2 from the autoclave.

The experimental conditions are listed in Table 1. The surface of the non-foamed samples was treated with acetic acid to see the phase morphology of the blended polymer. The phase morphology, as well as cellular structure, was observed by scanning probe microscopy (SPM, SPM-9500J3, Shimadzu) in the tapping mode with the cantilever OMCL-AC160TS-C2 (OLYMPUS) after dry-etching for 1 h by oxygen plasma (HDT-400, JEOL, Japan).

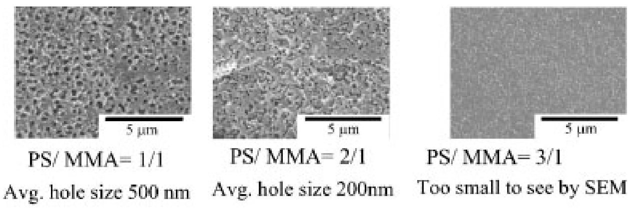

Figure 3. SEM micrographs of cross sectional area of samples treated by acetic acid before foaming.

Results and Discussion

Figure 2 shows the molecular weight distributions of the three polymer blends with different PS/PMMA ratios of 1/1, 2/1, and 3/1. In Figure 2, the solid line presents molecular weight distribution of neat PS and the open symbols represent molecular weight distributions of PS/PMMA blends with different PS/PMMA weight ratios. The molecular weight distribution was depicted by normalizing the data with the base and the highest point of its distribution. In the region lower than 1.2×10^5 $g \cdot mol^{-1}$, the differences between the solid line and symbols are small. This means that the molecular weight distribution of PS is dominant in this region. In the region higher than 1.2×10^5 $g \cdot mol^{-1}$, the difference becomes prominent and indicates the existence of polymerized MMA, i.e., PMMA. Subtracting the distribution profile of PS from each molecular weight distribution of PS/PMMA polymer blends, the weight averaged molecular weight of PMMA polymerized in PS was calculated to be 2.20×10^6 $g \cdot mol^{-1}$. For a reference, PMMA was simply polymerized with 1 wt.-% AIBN without using the PS matrix under the same temperature conditions in which polymer blends were made. The weight averaged molecular weight of the resulting PMMA was 1.50×10^5 $g \cdot mol^{-1}$. The PMMA polymerized in the PS matrix has a higher weight averaged molecular weight than that of PMMA made by bulk polymerization.

Figure 3 shows SEM micrographs of the cross sectional area of the polymerized samples treated with acetic acid. Only the PMMA domains in the PS matrix were removed by treatment with acetic acid and these domains were observed as holes. The sizes of the holes, i.e., the size of PMMA domains varied from 100 to 500 nm at PS/PMMA = 1/1. As the ratio of PS to PMMA increased, the average size of PMMA domains became smaller. At PS/PMMA = 3/1, the PMMA domain was invisible at SEM images even after dry-etching and rinsing with acetic acid.

The dispersive PMMA domains in PS and the difference between the plasticization effect of CO_2 on PMMA and PS could create nanoscale cell structures. Figure 4 shows the CO_2 induced reduction of glass transition temperatures of PMMA and PS.[8] As can be seen in Figure 4, when the CO_2 sorption pressure was 8.6 MPa, T_g of PMMA decreases to 0 °C while that for PS levels out at 30 °C. When foamed at 40 °C after dissolving CO_2 at 8.6 MPa, the bubble growth and coalescence were enhanced at both PMMA and PS phases because the foaming temperature was higher than the glass transition temperatures of both polymers. When foamed at 20 °C, bubble nucleation and growth were localized in PMMA domains because the foaming

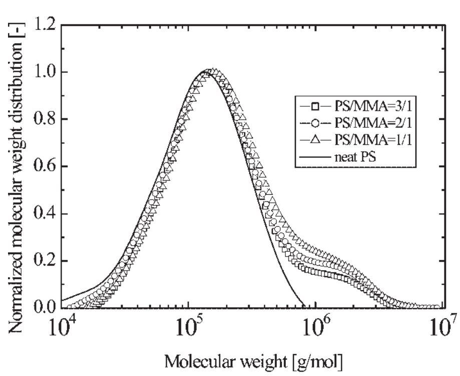

Figure 2. Molecular weight distribution of samples with different PS/MMA ratios. Solid line: neat PS; square: PS/MMA = 3/1; circle: PS/MMA = 2/1; triangle: PS/MMA = 1/1.

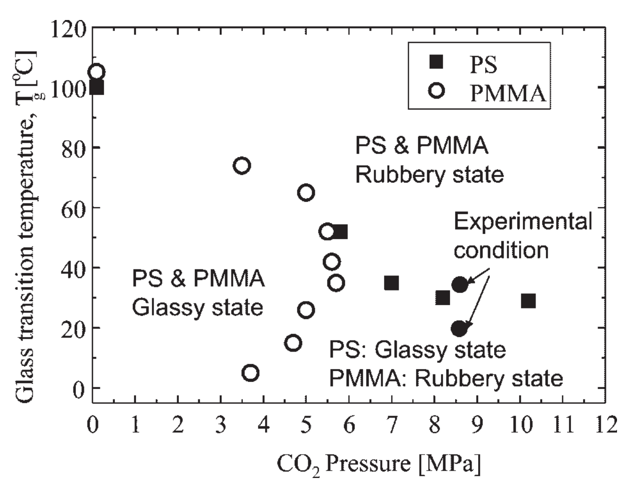

Figure 4. CO_2 induced T_g depression of PMMA and PS.

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temperature was lower than T_g of PS and the viscoelasticity of PS phase was high. The temperature effect on cell size around T_g agreed with the results of PS-*b*-PFMA nanocellular foam.[4]

Figure 5 (a) and (b) show the SPM images of samples before and after foaming. The foaming was conducted at 20 °C with a depressurization rate of 0.5 MPa · min^{-1}. The SPM height image shows the dark and bright regions corresponding to the valleys and hills of the surface. Figure 5 (b) shows the existence of voids (cells). The shape of cells is similar to the voids in Figure 3. The nanocells were created in the PMMA nanodomains by foaming of CO_2.

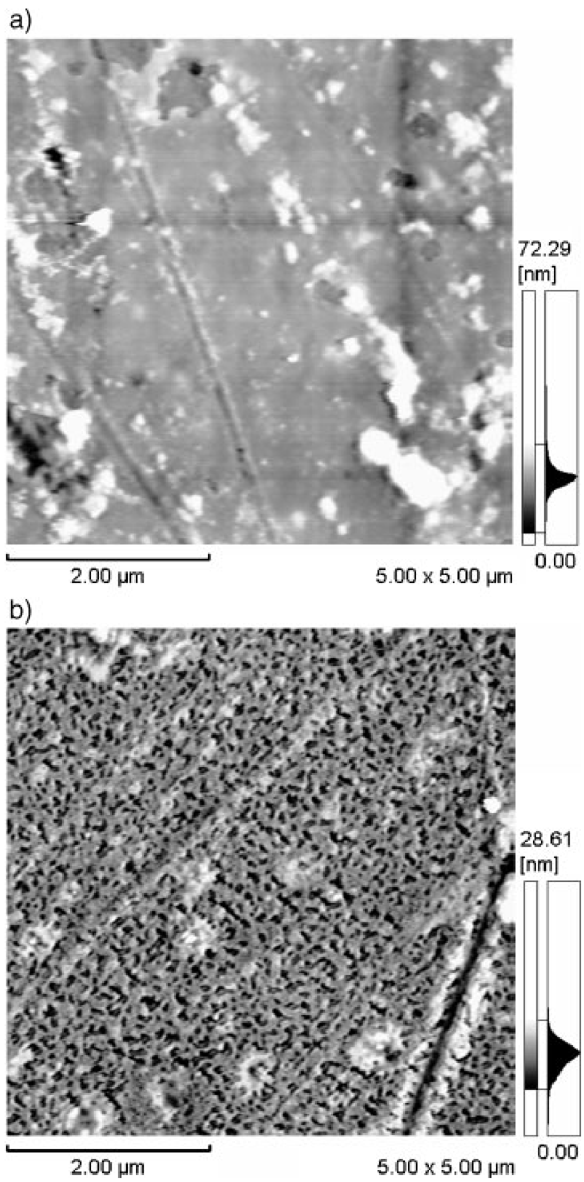

Figure 5. (a) SPM image of non-foamed sample after dry etching but before foaming. (b) Foamed sample at 20 °C with 0.5 MPa · min^{-1} after dry etching. PS/MMA = 2/1 for both (a) and (b).

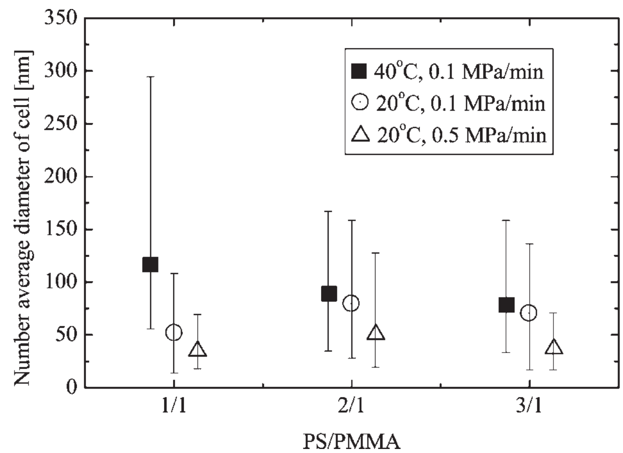

Figure 6. Effect of temperature and depressurization rate on the average diameter of a cell. Square: 40 °C, 0.1 MPa · min^{-1}; circle: 20 °C, 0.1 MPa · min^{-1}; triangle: 20 °C, 0.5 MPa · min^{-1}.

To study the effect of foaming temperature, foaming experiments were conducted at two different temperatures, 20 and 40 °C using a depressurization rate of 0.1 MPa · min^{-1}. When the samples were foamed at 40 °C with 0.5 MPa · min^{-1}, the cellular structure became microscale in size. Thus, the results of foaming under condition D in Figure 4 were excluded from the following analysis.

The average cell diameter and the density of cells were calculated from the SPM images of the samples foamed under conditions A, B and C. The calculation method as proposed by Kumar et al.[9] was employed in this study. The results are illustrated in Figure 6 and 7. Error bar indicates the maximum and minimum cell diameters observed at each foaming condition. When the depressurization rate was changed from 0.1 to 0.5 MPa · min^{-1}, the

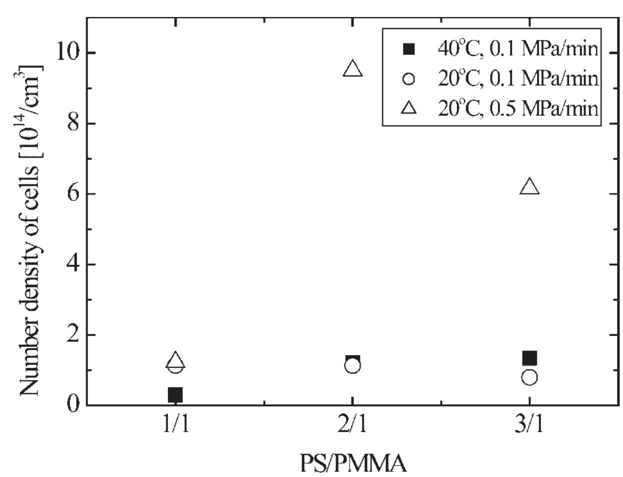

Figure 7. Effect of temperature and depressurization rate on cell density. Square: 40 °C, 0.1 MPa · min^{-1}; circle: 20 °C, 0.1 MPa · min^{-1}; triangle: 20 °C, 0.5 MPa · min^{-1}.

average cell diameter was reduced to 50 nm and cell density was increased at all samples of PS/MMA ratios. The rapid depressurization provides large degrees of super saturation. For polymer blends, the bubble nucleation was enhanced by a rapid pressure drop operation in two ways. It increases the number of bubbles in a PMMA domain as well as the number of PMMA domains in which the bubble was nucleated. The increase in the number of foamed PMMA domains could be prominent at the foaming of 2/1 and 3/1 PS/PMMA ratios, where the size of PMMA domain is small. This is likely the cause for the dramatic increase in cell density in 2/1 and 3/1 PS/PMMA samples when the depressurization rate was increased.

At the slower depressurization rate of 0.1 $MPa \cdot min^{-1}$, the difference in foaming temperatures at 40 °C and 20 °C, did not induce big differences in cell diameter and cell density for the samples of PS/MMA, 2/1 and 3/1. On the other hand, a PS/MMA ratio of 1/1, showed a large change in number average cell diameter as well as density when decreasing the foaming temperature from 40 to 20 °C. The disperse PMMA domain is larger as the PMMA weight ratio increases as shown in Figure 3. Thus, the bubble size in samples with a PS/PMMA ratio of 1/1 became larger as bubbles coalesced in the larger PMMA domain when temperature increased from 20 to 40 °C. On the other hand, for samples with PS/PMMA ratios of 2/1 and 3/1, the PMMA domain size in blend morphology was small enough to confine bubble growth and suppress bubble coalescence. Thus, an increase in temperature did not change cell size for samples with ratios of 3/1 and 2/1.

Conclusion

Well-understood methods of polymerization in a polymer matrix could provide highly dispersed PMMA domains in PS matrices. Nanocellular foam was created using a polymer blend and carbon dioxide by a pressure quench method. Bubble nucleation and cell size were controlled by adjusting foaming temperature, and blend ratio of PS/MMA. The concept of localizing bubble nucleation within a confined domain is a capable method for creating nanocellular foams from commodity polymer blends. The higher the depressurization rate was conducted for foaming, the smaller the bubble size and the larger the bubble density could be realized. Furthermore, there was an optimal PS/PMMA blend ratio to realize the finest bubble size with the highest bubble density. It was the nanocellular foams with 40–50 nm in average cell diameter and 8.5×10^{14} cm^{-3} in cell density.

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